Muir *et al. BMC Health Services Research* (2025) 25:709
https://doi.org/10.1186/s12913-025-12720-x

BMC Health Services Research

RESEARCH Open Access

Association of emergency department nurse and physician work environment agreement on clinician job and patient outcomes

K. Jane Muir[1,2,3*], Anish K. Agarwal[2,3,4], Daniela Golinelli[1], Karen B. Lasater[1,2], James Ballinghoff[4], Colleen Mattioni[5], Matthew D. McHugh[1,2], Linda H. Aiken[1,2] and United States Clinician Wellbeing Study Consortium

Abstract

Background Emergency medicine is a highly interdisciplinary field, and emergency nurses and physicians have high rates of burnout compared to other specialties. National and international agencies prioritize investments in systems-based solutions to improve clinicians' work environments. The objective of this study was to determine whether emergency department (ED) clinicians agree on the quality of work environments, and whether their agreement is associated with job outcomes, patient safety, and quality of care.

Methods This cross-sectional study used data from 1,604 ED nurses ($n=1,190$) and physicians ($n=414$) who completed the 2021 US Clinician Wellbeing Study in 47 Magnet hospitals. A K-means algorithm classified hospitals into 'profiles' based on nurse and physician agreement on work environment assessments. Hospital-level linear regression models determined the relationship between hospital profiles and clinician job and patient outcomes.

Results The overall clinician sample ($n=1,604$) was on average 39.4 years of age (SD = 11.2), 72.3% female, with 8.3 years of experience (SD = 7.9), 77.7% White, and 93.6% non-Hispanic. Two hospital profiles indicated clinician agreement: "Agree, Unfavorable Environment" ($n=10$ hospitals), and "Agree, Favorable Environment" ($n=15$); the third profile indicated disagreement: "Disagree, Less Favorable Environment among Nurses" ($n=22$). There were no hospital profiles with physicians rating their work environment less favorably than nurses. Compared to the "Agree, Favorable Environment" hospitals, the "Agree, Unfavorable Environment" and "Disagree, Less Favorable among Nurses" hospitals were associated with higher burnout (e.g., $\beta=25.8\%$, 95% CI 11.6, 40.1, $p<.001$ and $\beta=15.4$, 95% CI 3.7, 27.2, $p<.001$, respectively), job dissatisfaction, and intent to leave; and unfavorable patient care quality and unfavorable patient safety grades (e.g., $\beta=29.1\%$, 95% CI 18.4, 39.8, $p<.001$ and $\beta=11.9\%$, 95% CI 3.0, 20.8, $p<.01$, respectively).

Conclusions In this cross-sectional study, emergency nurses and physicians in almost half of study hospitals disagreed on the quality of the work environment, suggesting that two essential collaborators in high-stakes care do not agree on deficiencies in ED work environments. Sustainable systems-based solutions to improve ED work environments involve bridging these disparate workplace experiences.

Keywords Work environment, Burnout, Well-being, Emergency care

*Correspondence:
K. Jane Muir
janemuir@nursing.upenn.edu

Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Background

Nurses and physicians in hospital emergency departments (EDs) experience high rates of burnout compared to clinicians from any other health care specialty in the U.S [1, 2]. Given the unpredictable workloads and fast-paced, high stakes nature of EDs [3], the work environment is one of the few modifiable factors associated with both clinicians' job outcomes as well as the quality and safety of patient care [4]. Evidence demonstrates that when emergency nurses work in higher-quality work environments—where they have greater autonomy over their work, strong collegial relationships with physicians, supportive managers, and are involved in unit leadership— they have 61% lower odds of burnout, 67% lower odds of job dissatisfaction, and 54% lower odds of intentions to leave their employer within the year [5]. Poor physician work environment ratings are associated with higher physician burnout, while other evidence suggests that ED physician burnout is associated with patient adverse events [6] and prolonged ED wait times [7].

Physicians are unionizing at unprecedent rates alongside nurses who have organized for over two decades about the quality of their work environments [8]. These efforts signal concern that clinicians' perspectives about the quality of their work environment are insufficiently addressed by hospital employers. These workforce issues have become a priority policy and research concern identified by the U.S. Surgeon General and the National Academy of Medicine [9, 10].

The National Academy of Medicine's Action Collaborative on Clinician Well-Being [9] highlights the need for systems-based interventions to improve clinical work environments to reduce clinician burnout, improve well-being, and advance high-quality and safe patient care. A key emphasis of this national effort is to harmonize interventions that are inter-disciplinary, given that clinicians operate within teams, rather than siloes [11]. Emergency department care is described as being particularly inter-disciplinary, a feature that attracts many nurses and physicians to work there, as clinicians operate in team-based workflows physically co-located together in patient triage, trauma bays, and hallway care areas [12].

Despite the collaborative nature of the ED, the policies, strategies, and operational work flows to guide clinical care delivery and the cultivation of safe work environments in this setting are often siloed within medicine or nursing [13, 14]. Research siloed by clinician group may impede progress in advancing safe, high-quality care in the ED and ensuring that hospitals can recruit and retain clinicians in this setting [2, 15, 16].

To inform systems-based interventions that improve clinician well-being and patient care delivery in the ED, the objective of this study was to determine whether ED nurses and physicians agree on the quality of their hospital work environments and to evaluate whether the extent of agreement/disagreement was associated with clinician job outcomes, patient safety and quality of care. We hypothesized that we would identify four hospital profiles characterizing work environment agreement ((1) agree favorable, (2) agree unfavorable, (3) disagreement (nurses rate less favorably than physicians and (4) physicians rate less favorably than nurses). We also hypothesized that clinician agreement on better quality ED work environments would be associated with better clinician job and patient outcomes.

Conceptual model

The Social Ecological Model [17] is the conceptual model informing this study, which posits that a studied phenomenon is informed by inter-related individual and interpersonal factors, as well as larger systems factors. An individual clinician's experiences on the job and perceptions of patient care quality safety and quality can be influenced by the inter-disciplinary dynamics with other clinicians as well as systems factors such as their work environment (e.g., whether there is adequate staffing, strong physician-nurse relations, opportunities for clinicians to be involved in unit decision-making, clinician autonomy, supportive unit leadership). Prior qualitative investigations [3] have identified that, despite emergency nurses and physicians having distinct training and work responsibilities, their work is highly collaborative, suggesting that overlapping, inter-professional dynamics and systems factors are important to examine in the context of clinician well-being and patient care delivery.

Methods

Study design and data collection

The data for this cross-sectional study were initially collected from a multicenter collaborative Clinician Well Being Study of 60 Magnet hospitals [2] across 22 states, conducted through a common research protocol. Magnet hospitals have been designated by the American Nurses Credentialing Center in a formal organizational credentialing program that focuses on clinician well-being and retention and good patient outcomes; currently close to 10% of US hospitals are Magnet recognized [18, 19] The main study findings from the U.S. Clinician Well Being Study have been published elsewhere [2, 4]. The study was approved by the University of Pennsylvania Institutional Review Board.

The participating 60 U.S. Magnet hospitals sampled nurses and physicians across adult medical-surgical and ED settings via email to participate in the study. Data were collected from January to June 2021 by University of Pennsylvania researchers who managed all data collection and analysis. Data were anonymous and went directly to the research team. The overall clinician-level

survey response rate across the 60 participating hospitals was 26% [4], yielding responses from 21,000 clinicians total (n = 5,312 physicians and n= 15,738 nurses across all hospital units/specialties). The clinician-level response rate is consistent with large email-based surveys in recent years [20].

The primary interest of our study was not clinician-level responses, but rather organizational features of the hospitals derived from aggregated reports of clinicians working in the same hospitals. For this study hospitals were required to have a minimum of 5 nurse and 2 physician respondents working in their hospital ED. These were the minimum thresholds indicating acceptable agreement on the work environment for aggregation to the hospital level (intraclass correlation coefficient ≥ 0.60 [21]) in aggregating hospital-level measures of the exposure variable of interest—clinicians' work environments [22]. The mean number of nurse respondents per hospital was 22 and ranged from 5 to 69 nurses per hospital. The mean number of physician respondents was 8 and ranged from 2 to 31. The final sample included 1,604 ED clinicians (n = 1,190 nurses; n = 414 physicians) across 47 hospitals. There were no significant differences in the characteristics of those hospitals excluded (after dropping those not meeting the minimum threshold) compared to the included final sample (n = 47 hospitals, see Additional file 1).

Hospital characteristics

The clinician data were linked via a common hospital identifier with the 2019 American Hospital Association Annual Hospital Survey, which provided hospital-level characteristics including bed size, annual ED volume, technology capabilities, trauma hospital designation, urbanicity, and teaching status. Hospital size was defined as small (≤ 250beds), medium (251–500 beds), and large (> 500 beds). Annual ED volume was defined as low (< 40,000 patients per year), medium (40–80,000) and high (> 80,000). Teaching status was determined by the number of physician residents per bed and defined as non-teaching (no residents), minor (1:4 residents per bed), and major (> 1:4 residents per bed). Technology capabilities were defined as hospitals providing open-heart surgery and organ transplant services. Trauma designation was defined as hospital EDs with a certified trauma center.

Measures

The survey measures used in this data analysis were derived from a survey methodology published elsewhere and included the clinician work environment (exposure), clinician job outcomes (burnout, job dissatisfaction, intent to leave), and patient safety and quality care [22].

Work environment

The primary variable of interest was the clinician work environment, derived from the Clinician Well-Being survey. In the survey, nurses and physicians were asked to respond to the question, "Rate the overall quality of your work environment" by selecting one of the following choices: excellent, good, fair, poor. In statistical analyses, the variable was dichotomized to represent clinician reports as "good" and "poor". The measure was aggregated at the hospital level to report the proportion of nurses and physicians rating the work environment as poor. This measure has been used in previous research on clinician well-being demonstrating predictive validity [4].

Clinician job outcomes

The Clinician Well-Being survey provided measures of burnout, job dissatisfaction, and intent to leave employer as clinician job outcomes. Burnout was measured using the Maslach Burnout Inventory emotional exhaustion subscale, with responses over 27 indicating a score higher than the published top tertile of health care workers, or "high burnout" [4, 23]. A license to use this scale was provided to the author team (see Additional file 2). Clinicians responded to a single item question asking about job dissatisfaction using a 4-level Likert scale ranging from "Very satisfied" to "Very dissatisfied" [4, 22]. Finally, clinicians responded to a single item question about whether they intend to leave their hospital in the next year due to job dissatisfaction. Measures were dichotomized for the data analysis indicating high burnout, job dissatisfaction, and intent to leave the job.

Patient safety and quality care

Patient care quality and safety outcomes were also derived from the Clinician Well- Being survey. Clinicians reported the quality of patient care on the unit on a 4-point Likert scale that was dichotomized into "unfavorable quality of care" or "favorable quality of care". Clinicians provided a patient safety grade that ranged from A to F (as per grading school system with no E grade available; unfavorable scores were C, D, F), dichotomized into "unfavorable patient safety grade" and "favorable patient safety grade".

Data analysis

A K-means algorithm was used to identify the hospital profiles based on the nurse and physicians' work environment reports. K-means algorithms are a type of unsupervised machine learning that assign "similar hospitals" based on a specific data point—in this instance clinician work environment reports [24]. To determine the optimal number of clusters we used a variety of methods such as the elbow and silhouette method as implemented in the NbClust R package [24]. The package computes

30 different indices to determine the optimal number of clusters. In this case, the majority of these indices suggested that three clusters or hospital profiles were optimal.

Descriptive statistics were used to analyze characteristics of the hospital ED and clinician sample across the study outcomes. Race and ethnicity were self-reported by the clinicians in the survey and were included to inform the demographic diversity of the sample across clinician groups. Fisher's exact tests of significance, students t-tests, and analysis of variance (ANOVA) were used to measure differences in hospital characteristics (e.g., teaching status, ED volume) and clinician characteristics (e.g., age), across hospital profile type and clinician role.

Hospital-level linear regression models were constructed to determine the relationship between the hospital profiles and the percentage of hospital clinicians reporting job (burnout, job dissatisfaction, intent to leave) and patient care outcomes (unfavorable patient care quality, unfavorable safety grade) outcomes aggregated at the hospital level. Unadjusted models were first generated to determine the relationship between the hospital profiles and outcomes absent any controls, followed by models adjusted for hospital characteristics (e.g., teaching status, annual volume) and role (emergency nurse or physician). A linear model was used to produce predicted means of the study outcomes by clinician role (nurse or physician) and hospital profile.

Results

The final sample included 1,604 clinicians (n = 1,190 nurses; n = 414 physicians) employed in 47 U.S. Magnet hospitals (Table 1). On average, physicians were older than nurses (43 vs 38 years old, $p < 0.001$) with more years of experience (9.7 vs 7.7 years, $p < 0.001$), and less likely to be female (39.6% vs 86%, $p < 0.001$). Physicians were more likely to be non-Hispanic as compared to nurses (96% vs 93%, $p < 0.001$). Nurses compared to physicians had higher rates of burnout (52% vs 29%, $p < 0.001$), job dissatisfaction (24% vs 13%, $p < 0.001$), intent to leave (47% vs 17%, $p < 0.001$), and unfavorable patient care quality (17% vs 9%, $p < 0.001$) and safety grade (28% vs 15%, $p < 0.001$) reports.

Three hospital profiles were identified using the K-means algorithm (Fig. 1). Two hospital profiles indicated clinician agreement on unfavorable ("Agree, Unfavorable Environment; n = 10 hospitals) and favorable ("Agree, Favorable Environment; n = 15 hospitals) work environments; one hospital profile indicated clinician disagreement with nurses rating the work environment less favorable than physicians ("Disagree, Less Favorable among Nurses; n = 22 hospitals). No hospital profiles indicated physicians rating the work environment less favorably than nurses.

Characteristics of study hospitals across profiles are provided in Table 2. Hospitals where clinicians agreed on unfavorable work environments were on average larger (> 500 beds, p = 0.01) with trauma centers (80%, p = 0.03).

Table 1 Characteristics, job outcomes, and assessments of patient safety and care quality among emergency nurses and physicians in 47 Magnet hospitals

Characteristics	All Clinicians (n = 1,604)	RNs (n = 1,190)	MDs (n = 414)	p-value
Overall Sample Demographics				
Age (years), mean (SD)	39.4 (11.2)	38.0 (11.0)	43.3 (10.8)	<.001
Female, No. (%)	1,099 (72.3)	943 (**83.7**)	156 (**39.6**)	<.001
Years in Hospital, mean (SD)	8.3 (7.9)	7.7 (7.5)	9.7 (8.9)	<.001
Ethnicity				
Non-Hispanic, No (%)	1,415 (93.6)	1,039 (92.8)	376 (95.9)	0.03
Race[*]				
Asian	243 (8.9)	86 (7.2)	57 (13.8)	NA
American Indian or Alaskan Native	14 (0.9)	13 (1.1)	NR	NA
Black or African American	58 (3.6)	47 (3.9)	11 (2.7)	NA
Native Hawaiian or Other Pacific Islander	13 (0.8)	12 (1.0)	NR	NA
Other	67 (4.2)	54 (4.5)	13 (3.1)	NA
White	1,246 (77.7)	937 (78.7)	309 (74.6)	NA
Job Outcomes				
High burnout, n (%)	733 (46.2)	612 (52.1)	121 (29.4)	<.001
Job dissatisfaction, n (%)	338 (21.2)	284 (24.0)	54 (13.1)	<.001
Intent to leave, n (%)	606 (38.7)	536 (46.6)	70 (17.0)	<.001
Patient Outcomes				
Poor patient care quality, n (%)	235 (14.7)	199 (16.8)	36 (8.7)	<.001
Unfavorable patient safety grade, n (%)	397 (24.8)	337 (28.4)	60 (14.5)	<.001

Abbreviations: MD Physician, *RN* Registered nurse, *SD* Standard deviation, *NA* Not applicable, *NR* Not reported due to small cell size

[*]Clinicians could select all that apply for race categories, thus percentages will not equal 100; Values may not add up to 100 due to missingness

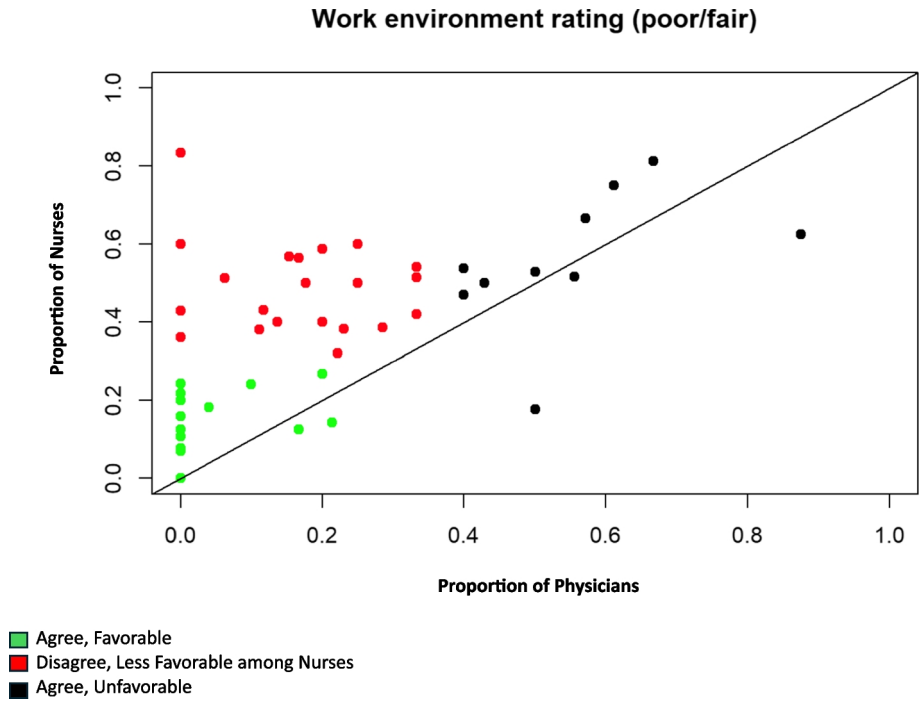

Fig. 1 K-means clusters identifying hospital profiles of clinician work environment agreement. Note. Green values indicate "Agree, Favorable Environment"; Red values indicate "Disagree, Less Favorable for Nurses"; Black values indicate "Agree, Unfavorable Environment" using the proportion of clinicians rating their work environment as "poor or fair" from the Clinician WellBeing Study of emergency nurses and physicians

Table 2 Hospital characteristics in 47 study hospitals across hospital profiles

	Hospital Profile			
	Agree, Unfavorable Environment (*n* = 10)	**Agree, Favorable Environment (*n* = 15)**	**Disagree, Less Favorable among Nurses (*n* = 22)**	
Hospital characteristic				*p-value*
Hospital size, *n* (%)				0.01
≤ 250	2 (20.0)	8 (53.3)	1 (4.6)	
251–500	3 (30.0)	2 (13.3)	10 (45.4)	
> 500	5 (50.0)	5 (33.4)	11 (50.0)	
Annual ED Patient Volume, *n*(%)				0.08
0–40,000	0 (0.0)	7 (46.7)	3 (13.6)	
40–80,000	6 (60.0)	5 (33.3)	11 (50.0)	
> 80,000	4 (40.0)	3 (20.0)	8 (36.4)	
Teaching status, *n* (%)				0.50
No	3 (30.0)	7 (46.7)	5 (22.7)	
Minor	4 (40.0)	4 (26.7)	6 (27.3)	
Major	3 (30.0)	4 (26.6)	11 (50.0)	
Technology status, *n* (%)				0.16
Not	2 (20.0)	7 (46.7)	4 (18.2)	
High	8 (80.0)	8 (53.3)	18 (81.8)	
Trauma Center, *n* (%)				0.03
No	1 (10.0)	9 (60.0)	7 (31.8)	
Yes	9 (90.0)	6 (40.0)	15 (68.2)	
Poor Work Environment, %				<.001
% RNs	55.8	14.3	49.2	
% MDs	55.0	4.8	16.2	

Fisher's exact tests were performed for categorical variables (ED annual volume, teaching status, technology status, trauma center) and global analyses of variance (ANOVA) tests were performed for continuous variables (Avg. % of clinicians rating their work environment as poor)

Hospitals with clinician agreement on favorable work environments were more likely to be smaller (≤ 250 beds, $p = 0.01$) and non-trauma centers (60%, $p = 0.03$), while hospitals with clinician disagreement (less favorable among nurses) were on average larger (> 500 beds, $p = 0.01$) trauma centers (68.2%, $p = 0.03$).

Hospital-level linear regression models (Table 3), unadjusted and adjusted for hospital characteristics and clinician role identified a statistically significant relationship between the hospital clusters and all study outcomes. Compared to hospitals with clinician agreement on a favorable work environment ("Agree, Favorable Environment"), hospitals characterized as "Agree, Unfavorable Environment" and "Disagree, Less Favorable among Nurses" were associated with increased clinician burnout, job dissatisfaction, intent to leave, and unfavorable patient care quality and safety grades in the adjusted models. Clinician role was significantly associated with all study outcomes such that higher percentages of burnout were observed among nurses as compared to physicians (β = 21.4%, 95% CI 13.3, 29.4, $p < 0.001$), as well as job dissatisfaction (β = 12.2%, 95% CI 7.3, 17.0, $p < 0.001$), intent to leave (β = 29.5%, 95% CI 22.7, 36.2, $p < 0.001$), and unfavorable patient care quality (β = 9.3%, 95% CI 4.9, 13.7, $p < 0.001$) and safety grades (β = 15.9%, 95% CI 9.6, 22.1, $p < 0.001$).

In the adjusted linear regression models, all-clinician burnout in hospitals with clinicians' agreement on unfavorable work environments ("Agree, Unfavorable Environment") was 25.8 percentage points higher when compared to hospitals where clinicians agreed favorably on their environment ("Agree, Favorable Environment") (β = 25.8, 95% CI 11.6, 40.1, $p < 0.001$). All-clinician burnout in hospitals with nurses rating their environment less favorably than physicians was 15.4 percentage points higher (β = 15.4%, 95% CI 3.7, 27.2, $p < 0.001$), when compared to the "Agree, Favorable" hospitals.

The predicted means of the study outcomes by role and hospital profile are provided in Table 4. The rates of nurse job and patient care outcomes were significantly higher in

Table 3 Hospital-level linear regression models determining the relationship between hospital profiles and clinician job and patient care quality and safety outcomes

	Beta Coefficient Unadjusted (95% CI)	Beta Coefficient Adjusted for Hospital Characteristics (95% CI)
Burnout		
Agree, Favorable Environment (ref)	–––	–––
Agree, Unfavorable Environment	22.6*** (11.3, 33.9)	25.8*** (11.6, 40.1)
Disagree, Less Favorable among Nurses	16.1** (6.8, 25.3)	15.4** (3.7, 27.2)
Nurse (reference: physician)	21.4*** (13.3, 29.4)	21.4* (13.1, 27.2)
Job Dissatisfaction		
Agree, Favorable Environment (ref)	–––	–––
Agree, Unfavorable Environment	30.4*** (23.7, 37.2)	32.5*** (24.2, 40.9)
Disagree, Less Favorable among Nurses	11.7** (6.2, 17.3)	12.6*** (5.7, 19.5)
Nurse (reference: physician)	12.2*** (7.4, 17.0)	12.2*** (7.3,17.0)
Intent to Leave		
Agree, Favorable Environment (ref)	–––	–––
Agree, Unfavorable Environment	30.3*** (21.0, 39.6)	31.7*** (20.0, 42.3)
Disagree, Less Favorable among Nurses	11.8** (4.2, 19.4)	12.7* (3.1, 22.3)
Nurse (reference: physician)	29.5*** (22.8, 36.1)	29.5*** (22.7, 36.2)
Unfavorable patient care quality		
Agree, Favorable Environment (ref)	–––	–––
Agree, Unfavorable Environment	18.7*** (12.5, 24.9)	18.4*** (10.8, 25.9)
Disagree, Less Favorable among Nurses	7.5** (2.4, 12.6)	6.26* (2.7, 12.5)
Nurse (reference: physician)	9.3** (4.8, 12.6)	9.3*** (4.9, 13.7)
Unfavorable patient safety grade		
Agree, Favorable Environment (ref)	–––	–––
Agree, Unfavorable Environment	27.6*** (18.7, 36.5)	29.1*** (18.4, 39.8)
Disagree, Less Favorable among Nurses	11.7** (4.4, 19.1)	11.9** (3.0, 20.8)
Nurse (reference: physician)	15.9*** (9.5, 22.2)	15.9*** (9.6, 22.1)

CI Confidence interval. Models represent percentage increases and are adjusted for hospital characteristics of hospital bed size, annual ED volume, technology capabilities, and teaching status

*Indicates significance level of $p < .05$

**Indicates significance level of $p < .01$

***Indicates significance level of $p < .001$

Table 4 Clinician job and patient outcomes predicted means by role and across hospital profiles

	Agree, Favorable Environment (*n* = 15 hospitals)	Disagree, Less Favorable among Nurses (*n* = 22 hospitals)	Agree, unfavorable Environment (*n* = 10 hospitals)
Burnout			
Nurse	38.7	54.1**	64.5***
Physician	17.3	32.7	43.1
Job Dissatisfaction			
Nurse	10.9	23.5**	43.4***
Physician	– 1.3§	11.3*	31.2***
Intent to Leave			
Nurse	34.1	46.8*	65.7**
Physician	4.6	17.3	36.3**
Poor Care Quality			
Nurse	9.9	16.2***	28.3***
Physician	0.6	6.9	19.1
Unfavorable patient safety			
Nurse	16.7	28.6**	45.8***
Physician	0.8	12.8	29.9*

Expected values represent the percentage of clinicians reporting the outcome

*Indicates significance level of $p < .05$

**Indicates significance level of $p < .01$

***Indicates significance level of $p < .001$

§The negative value for physicians is due the low rate of job dissatisfaction among physicians working in hospitals in the second cluster. Both the linear and normal assumptions might not be met in this case

the "Agree, Unfavorable Environment" compared to the "Agree, Favorable Environment" hospitals. In these hospitals where clinicians agreed on an unfavorable environment, all outcomes except for burnout and patient care quality were statistically higher for physicians, as compared to the favorable agreement hospitals. In the hospitals where clinicians disagreed ("Disagree, Less Favorable Among Nurses"), the rates of all clinician job and patient outcomes were higher for nurses as compared to the favorable agreement hospitals. Job dissatisfaction was the only outcome for physicians that was significantly higher in the disagreement hospitals as compared to favorable agreement hospitals. Physician burnout, intent to leave, patient care quality, and patient safety grade were higher, although not statistically different, in the disagreement compared to favorable agreement hospitals.

Limitations

This study included a sample of Magnet hospitals only – which are known to have favorable work environments [19, 25]; nevertheless, we observed variation in the degree of ED physician and nurse agreement on the work environment. Our findings may represent an underestimation of the variation in clinician work environment agreement among all hospitals, since Magnet hospitals are known for good physician and nurse collaboration [26]. Clinician demographics in our sample may not be representative of all ED nurses and physicians given our sampling of Magnet hospitals. Our regression models are based on observational, cross-sectional data and thus causal inference cannot be applied to our findings. Our sample included higher numbers of nurses than physicians, which reflects the appropriate proportion of nurses and physicians employed in a typical hospital ED [3]. The study included reports from clinicians on patient outcomes. Prior studies demonstrate that clinician reports on patient care quality and safety are correlated with objective patient outcomes, which is consistent with the key role that clinicians have in bedside surveillance and care delivery [27].

Discussion

This study is one of the first to identify the variation in nurse and physician agreement on the work environment across 47 hospital EDs. Examining the perspectives across emergency nurses and physicians is critically important as inter-professional collaboration and team-based care is the pillar of clinical workflow and patient outcomes in acute care delivery. In the fast-paced world of hospital EDs where nurses and physicians work closely together to save lives, there is an unexpectedly high degree of disagreement between them about the quality of the work environment in some hospitals—with nurses rating ED work environments much worse than physicians in general. Our study is unique in identifying a lack of agreement between emergency nurses and physicians and how it varies across hospitals. That variation was consequential and significantly associated with outcomes including worse clinician well-being for both groups as well as heightened concerns about quality and safety of patient care. The implication of these findings is that, if two essential partners in emergency care within the same institution do not agree on the deficiencies in ED work environments, significant interdisciplinary research is needed to bridge these gaps and disparate experiences. The findings also provide important information in the context of rising clinician burnout and job turnover [2, 3, 5] that is exacerbated in hospital EDs struggling to recruit and retain clinicians [15].

Of the 47 study hospitals, clinicians in roughly half of hospitals disagreed on the favorability of their work environments. In all cases, nurses rated their environment less favorably than their physician colleagues. We did not identify a hospital profile where physicians rated their environment less favorably than nurses, suggesting that a prevailing workforce concern in hospital EDs is the work experience of nurses specifically who comprise the largest clinician group in these settings.

In all hospital-level regression models in this study, hospital profiles with unfavorable work environment

ratings among either all clinicians or nurses were associated with increases in all-clinician burnout, job dissatisfaction, intent to leave, and unfavorable patient quality and patient safety. In other words, even in hospitals with physicians rating their environment more favorably than nurses—these profiles were associated with poor clinician and patient care quality and safety outcomes compared to hospitals when both physicians and nurses agreed on a favorable work environment. This suggests that the presence of just one clinician groups' (e.g., nurses') lack of adequate resources, leadership support, and/or autonomy in the work environment may detrimentally impact or have a "spill over" effect on the overall quality and safety of patient care delivery and all clinician job outcomes in the ED. Our findings also suggest that outcomes by clinician role were associated with work environment disagreement. For example, physician rates of job dissatisfaction were significantly higher in hospitals where nurses rated their environment less favorable than physicians, as compared to hospitals where both clinicians agreed on a favorable environment.

The relationship between ED nurses and physicians is uniquely interprofessional [28]. Yet, close to half of our study hospitals reported disagreement between emergency nurses and physicians on their work environment ratings, with nurses rating less favorably than physicians. The extent to which ED physicians and nurses agree on a favorable work environment, as our findings demonstrate, is associated with the quality and safety of patient care delivery and the well-being outcomes of clinicians in a high-stress setting. To address gaps in agreement, our findings suggest that greater investment in improving nurses' work environments is needed to strengthen care delivery and clinician well-being in the ED. Specific approaches to address nurses' work environments in the ED include ensuring that nurses are staffed safely in their workplace, improving nurse autonomy (e.g., establishing care protocols and algorithms for nurses to advance care in the ED) [12, 29], increasing nurse involvement in hospital affairs (i.e., regarding care delivery or policy changes in the ED), providing nurses with supportive managers, and ensuring nurse-physician collegiality (e.g., training clinicians in trauma patient care stabilization as a team rather than in siloes).

The work environment is a modifiable feature of hospitals [29], and evidence-based practices that align across emergency nurses and physicians are needed to improve EDs to address disparities in work environment experiences. In clinical practice, hospital administrators should consider how, despite having distinct roles in ED settings, nurse and physicians' agreement on a favorable work environment experience supports cohesive patent care delivery that inform patients' and clinicians' outcomes. In a larger study of nurses and physicians working in Magnet hospitals [4], both physicians and nurses agreed that improving a key feature of nurses' work environment—nurse staffing levels—is the intervention that they would prioritize to reduce their burnout and improve well-being. A large number of studies in the literature demonstrates an association between safer nurse staffing levels and reductions in nurse burnout, job dissatisfaction and intent to leave [30, 31]. From a policy perspective, physicians can support legislative efforts [32] that would mandate that hospitals staff nurses adequately in EDs, given the empirical relationship between nurse staffing levels and patient outcomes [30]; and that nurse staffing levels may impact physicians' job outcomes [4].

That physicians prioritize improving nurse staffing levels as an evidence-based strategy to foster a safer work environment for themselves validates that physicians and nurses need adequately resourced working conditions to deliver high-quality care and remain working in their jobs. Additional interventions rated highly by both included reduced documentation burden and management responsive to clinician concerns [4]. Taken together, our results highlight opportunities for greater interdisciplinary collaboration around work environment improvement initiatives as well as efforts to create a common understanding of each group's experience of their work environment.

Conclusions

This cross-sectional study found that ED nurses and physicians in close to half of study hospitals disagreed on the quality of their work environment. Clinician job and patient outcomes were worse when both clinicians rated their work environment unfavorably, and when nurses only reported their work environments as unfavorable. Health care leaders must prioritize improvements in work environments aligned across clinician groups to cultivate clinician well-being and advance safer care delivery.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12913-025-12720-x.

Supplementary Material 1.

Supplementary Material 2.

Acknowledgements

The authors thank the American Nurses Credentialing Center (ANCC) for encouragement and administrative support to the US Clinician Wellbeing Consortium and the Magnet4Europe Consortium (www.magnet4europe.eu) funded by European Union's Horizon 2020 research program (grant 848031). The authors also thank Tim Cheney for his statistical support.

U.S. Clinician Wellbeing Study Consortium

Directors: LH Aiken and MD McHugh, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA. Advocate Christ Medical Center, Oak Lawn, IL, M Cleary, PI; Anne Arundel Medical Center, Annapolis, MD, C Ley, PI; Avera McKennan Hospital & University Health Center, Sioux Falls, SD, CJ Borchardt, PI; Billings Clinic, Billings, MT, JM Brant, PI; Bon Secours/St Elizabeth Youngstown Hospital, Youngstown, OH, BL Turner, PI; Bon Secours/St Mary's Hospital, Richmond, VA, AE Leimberger, PI; Bristol Health, Bristol, CT, K Waterman, PI; Cedars-Sinai Medical Center, Los Angeles, CA, BL Coleman, PI; Cleveland Clinic Foundation, Cleveland, OH, NM Albert, PI; El Camino Health, Mountain View, CA, C Stewart, PI; Emory Saint Joseph's Hospital, Atlanta, GA, D Steele, PI; Emory University Hospital, Atlanta, GA, R Kaplow, PI; Englewood Health, Englewood, NJ, K Kaminsky, PI; Good Samaritan Hospital, Vincennes, IN, HA Hinkle, PI; Hackensack/Jersey Shore Medical Center, RD Besa, PI; Hackensack/Raritan Bay Medical Center & Old Bridge Medical Center, Perth Amboy, NJ, KP Taylor, PI; Hackensack/Riverview Medical Center, R Graboso, PI; Hackensack/University Medical Center, Hackensack, NJ, K Dimino, PI; Huntington Hospital, Pasadena, CA, L Searle-Leach, PI; Kootenai Health, Coeur D Alene, ID, S Albritton, PI; Lehigh Valley Hospital, Allentown, PA, CL Davidson, PI; Loyola University Medical Center, Maywood, IL, T Carrigan, PI; Massachusetts General Hospital, Boston, MA, DA Burke, PI; Mid Coast Hospital, Brunswick, ME, KR Anthony, PI; Morristown Medical Center, Morristown NJ, MO Kowalski, PI; Newport Hospital, Newport, RI, M Rounds, PI; NorthBay Healthcare, Fairfield, CA, JM Tudor, PI; Northwell/Huntington Hospital, Huntington, NY, L Griffis, PI; Northwell/Long Island Jewish Medical Center, New Hyde Park, NY, LM Vassallo, PI; Northwell/Mather Hospital, Port Jefferson, NY, M Mulligan, PI; Northwell/North Shore Medical Center, I Macyk, PI; Northwell/Northern Westchester Hospital, Mount Kisco, NY, C Manley-Cullen, PI; Northwestern/Central DuPage Hospital, Winfield, IL, AE Haberman, PI; Northwestern/Delnor Hospital, Geneva, IL, SL Hutchinson, PI; Northwestern/Lake Forest Hospital, Lake Forest, IL, AL Barnard, PI; Northwestern Memorial Hospital, Chicago, IL, B Gobel, PI; Ohio State University Comprehensive Cancer Center, The James, Hilliard, OH, DL McMahon, PI; OSF Healthcare/Saint Anthony Medical Center, Rockford, IL, MJ Brown, PI; OSF Healthcare/Saint Joseph Medical Center, Bloomington, IL, L Strack, PI; OSF Healthcare/Saint Francis Medical Center, Peoria, IL, SA Emmerling, PI; Penn/Chester County Hospital, West Chester, PA, AR Coladonato, PI; Penn/Hospital of the University of Pennsylvania, Philadelphia, PA, JA Reich, PI; Penn/Lancaster General Hospital, Lancaster, PA, JJ Gavaghan, PI; Penn/Pennsylvania Hospital, Philadelphia, PA, FD Vanek, PI; Penn/Presbyterian Medical Center, Philadelphia PA, JR Ballinghoff, PI; Penn/Princeton Health, Plainsboro Township, NJ, KA Book, PI; Robert Wood Johnson University Hospital, New Brunswick, NJ, K Easter, PI; Southwestern Vermont Health Care, Bennington, VT, PB Seaman, PI; Stanford Health Care, Stanford, CA, ME Lough, PI; Summa Health, Akron, OH, CL Benson, PI; The Miriam Hospital, Providence, RI, Maria Ducharme, PI; The Valley Hospital, Ridgewood, NJ, Paul Quinn, PI; Thomas Jefferson University Hospital, Philadelphia, PA, DM Molyneaux, PI; UC Davis, Sacramento, CA, L Kennedy Madden; PI; UC Irvine Health, Orange, CA, D Grochow, PI; UC San Diego Health, San Diego, CA, E Nyheim, PI; University of Alabama at Birmingham Hospital, Birmingham AL, SM Purcell, PI; University of Iowa Hospitals & Clinics, Iowa City, IA, K Hanrahan, PI; University of Kentucky HealthCare, Lexington, KY, KB Isaacs, PI; WakeMed Health and Hospitals, Raleigh, NC, JJ Whade, PI.

Authors' contributions

KJM, AKA, MM, and LHA conceived and designed the study. KJM AKA and LHA obtained research funding. MM, KBL, LHA, DG the U.S. Clinician Wellbeing Consortium, JB, CM, and AKA supervised the conduct of the study and data collection. KJM drafted the manuscript, and all authors contributed substantially to its revision. KJM takes responsibility for the paper as a whole.

Funding

This research was funded by the Clinician Well-being Study Consortium and the National Institutes of Health (R01 NR014855 and T32 NR007104). Support was also provided by the Agency for Healthcare Research and Quality AHRQ R01HS028978 (K B Lasater), the National Institute of Nursing Research 1K01NR021419 (Muir), and the National Clinician Scholars Program and the Emergency Medicine Foundation/Emergency Nurses Association Foundation (K J Muir and A K Agarwal).

Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and anonymity of the sample. The data are stored at the University of Pennsylvania.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the University of Pennsylvania Institutional Reviewer Board. Participants provided written informed consent before initiating the survey informing the data for the study analysis.

Consent for publication

N/A.

Competing interests

The authors declare no competing interests.

Author details

[1]Center for Health Outcomes and Policy Research, School of Nursing, University of Pennsylvania, 418 Curie Boulevard, Fagin Hall, Philadelphia, PA 19104, USA
[2]The Leonard Davis Institute of Health Economics, University of Pennsylvania, Philadelphia, PA, USA
[3]Department of Emergency Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA
[4]Penn Presbyterian Medical Center and UPHS, Philadelphia, PA, USA
[5]Hospital of the University of Pennsylvania, Philadelphia, PA, USA

Received: 15 August 2024 / Accepted: 8 April 2025
Published online: 16 May 2025

References

1. Melnikow J, Padovani A, Miller M. Frontline physician burnout during the COVID-19 pandemic: national survey findings. BMC Health Serv Res. 2022;22(1):365.
2. Turnbach E, Coates L, Vanek FD, et al. Emergency nurses' well-being in Magnet hospitals and recommendations for improvements in work environments: A multicenter cross-sectional observational study. J Emerg Nurs. Published online July 2023. https://doi.org/10.1016/j.jen.2023.06.012.
3. Muir KJ, Keim-Malpass J, LeBaron VT. "You have to ask yourself when you've had enough": An ethnography of multi-level nurse burnout cultural impacts in the emergency department. SSM-Qualitative Research in Health. 2022;2:100111.
4. Aiken LH, Lasater KB, Sloane DM, et al. Physician and nurse well-being and preferred interventions to address burnout in hospital practice: Factors associated with turnover, outcomes, and patient safety. JAMA Health Forum. 2023;4(7):e231809. https://doi.org/10.1001/jamahealthforum.2023.1809.
5. Smalley CM, Simon EL, Meldon SW, et al. The impact of hospital boarding on the emergency department waiting room. JACEP Open. 2020;1(5):1052–9. https://doi.org/10.1002/EMP2.12100.
6. Lu DW, Dresden S, McCloskey C, Branzetti J, Gisondi MA. Impact of burnout on self-reported patient care among emergency physicians. Western Journal of Emergency Medicine. 2015;16(7):996.
7. De Stefano C, Philippon AL, Krastinova E, et al. Effect of emergency physician burnout on patient waiting times. Intern Emerg Med. 2018;13:421–8.
8. Ahmed AM, Kadakia K, Ahmed A, Shultz B, Li X. Trends in labor unionization among US health care workers, 2009–2021. JAMA. 2022;328(24):2404–11.
9. National Academies of Sciences and Medicine. Taking action against clinician burnout: a systems approach to professional well-being. 2019.
10. Office of the Surgeon General (OSG). Addressing Health Worker Burnout: The U.S. Surgeon General's Advisory on Building a Thriving Health Workforce. Washington (DC): US Department of Health and Human Services; 2022. Available from: https://www.ncbi.nlm.nih.gov/books/NBK595228/.
11. Chang BP, Cato K. Tackling burnout with team science: nursing and physician collaborations on improving psychological well-being among emergency clinicians. J Emerg Nurs. 2020;46(5):557–9.
12. Muir KJ, Keim-Malpass J, LeBaron VT. Examining the cultural impacts of an emergency department move using ethnography. Int Emerg Nurs. 2021;59:101082. https://doi.org/10.1016/J.IENJ.2021.101082.
13. Gillespie GL, Jiggens L, Russe C, Jinno L, Higham M, Meyer HA. Defining research funding priorities: Emergency Nurses Association Foundation. J Emerg Nurs: Published online; 2024.
14. Crilly J, Huang YL, Krahe M, et al. Research priority setting in emergency care: A scoping review. JACEP Open. 2022;3(6). https://doi.org/10.1002/emp2.12852.

15. Preiksaitis C, Krzyzaniak S, Bowers K, et al. Characteristics of emergency medicine residency programs with unfilled positions in the 2023 Match. Ann Emerg Med. Published online 2023. https://doi.org/10.1016/j.annemergmed.2023.06.002.
16. Muir KJ, Sloane DM, Aiken LH, Hovsepian V, McHugh MD. The association of the emergency department work environment on patient care and nurse job outcomes. J Am Coll Emerg Physicians Open. 2023;4(5):e13040.
17. Kilanowski JF. Breadth of the Socio-Ecological Model. J Agromedicine. 2017;22(4):295–7. https://doi.org/10.1080/1059924X.2017.1358971.
18. Kutney-Lee A, Stimpfel AW, Sloane DM, Cimiotti JP, Quinn LW, Aiken LH. Changes in patient and nurse outcomes associated with magnet hospital recognition. Med Care. 2015;53(6):550.
19. Kelly LA, McHugh MD, Aiken LH. Nurse outcomes in Magnet® and non-Magnet hospitals. J Nurs Admin. 2011;41(10). https://doi.org/10.1097/NNA.0b013e31822eddbc
20. Wu MJ, Zhao K, Fils-Aime F. Response rates of online surveys in published research: A meta-analysis. Computers in Human Behavior Reports. 2022;7:100206.
21. Estabrooks CA, Midodzi WK, Cummings GG, Ricker KL, Giovannetti P. The impact of hospital nursing characteristics on 30-day mortality. JONA: J Nurs Admin. 2011;41(7/8):S58-68.
22. Lasater KB, Jarrín OF, Aiken LH, McHugh MD, Sloane DM, Smith HL. A methodology for studying organizational performance: A multistate survey of front-line providers. Med Care. 2019;57(9). https://doi.org/10.1097/MLR.0000000000001167.
23. Maslach C. A multidimensional theory of burnout. Theories of organizational stress. 1999;(August):68–85.
24. Charrad M, Ghazzali N, Boiteau V, Niknafs A. NbClust: an R package for determining the relevant number of clusters in a data set. J Stat Softw. 2014;61:1–36.
25. McHugh MD, Kelly LA, Smith HL, Wu ES, Vanak JM, Aiken LH. Lower mortality in magnet hospitals. Med Care. 2013;51(5). https://doi.org/10.1097/MLR.0b013e3182726cc5.
26. Sermeus W, Aiken LH, Ball J, et al. A workplace organisational intervention to improve hospital nurses' and physicians' mental health: study protocol for the Magnet4Europe wait list cluster randomised controlled trial. BMJ Open. 2022;12(7):e059159. https://doi.org/10.1136/BMJOPEN-2021-059159.
27. McHugh MD, Stimpfel AW. Nurse reported quality of care: A measure of hospital quality. Res Nurs Health. 2012;35(6):566–75. https://doi.org/10.1002/nur.21503.
28. Fry M. An ethnography: Understanding emergency nursing practice belief systems. Int Emerg Nurs. 2012;20(3):120–5. https://doi.org/10.1016/j.ienj.2011.09.002.
29. Lake ET. Development of the practice environment scale of the nursing work index. Res Nurs Health. 2002;25(3):176–88. https://doi.org/10.1002/nur.10032.
30. Aiken LH, Clarke SP, Sloane DM, Sochalski J, Silber JH. Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction. J Am Med Assoc. 2002;288(16):1987–93. https://doi.org/10.1001/jama.288.16.1987.
31. McHugh MD, Rochman MF, Sloane DM, Berg RA, Mancini ME, Nadkarni VM, Merchant RM, Aiken LH; American Heart Association's Get With The Guidelines-Resuscitation Investigators. Better nurse staffing and nurse work environments associated with increased survival of in-hospital cardiac arrest patients. Medical care. 2016;54(1):74–80.
32. Dierkes A, Do D, Morin H, Rochman M, Sloane D, McHugh M. The impact of California's staffing mandate and the economic recession on registered nurse staffing levels: A longitudinal analysis. Nurs Outlook. 2022;70(2):219–27. https://doi.org/10.1016/J.OUTLOOK.2021.09.007.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.